Prismoidal and Classical Solid Volume Calculator

A self-contained Python/Tkinter graphical calculator for classical solid volumes, prismoids, prismatoids, frusta, average end-area volumes, general prismoidal corrections, and homothetic irregular-polygon frusta.

The application is intended for technical geometry, surveying, earthworks, engineering volume checks, and teaching/verification of prismoidal and frustum formulae. It combines numerical volume calculation with a large 3D graphical view of the selected solid and a full text calculation report.

Main capabilities

- Graphical fixed-size Tkinter interface.
- Left panel for solid selection, input data, calculated volume, correction line, and full calculation report.
- Right panel with a large Matplotlib 3D view of the selected solid.
- Dimension labels in the 3D view.
- Output report displayed directly in the GUI.
- Report export to `output.txt` using the **Save TXT** button.
- Decimal comma accepted in numeric inputs.
- Input validation for positive lengths, areas, and radii.
- Built-in usage/build instructions accessible from the GUI.
- Compatible with normal Python execution and PyInstaller one-file executable builds.

Supported solids and calculation methods

The script includes the following solids and volume methods:

1. Cube
2. Triangular prism
3. Regular hexagonal prism
4. Oblique prism / parallelepiped
5. Right rectangular prism / cuboid
6. Cylinder
7. Square pyramid
8. Right circular cone
9. Frustum of a square pyramid / similar-base frustum
10. Frustum of a cone
11. Triangular wedge
12. General prismoid / prismatoid
13. Sphere
14. Average end-area method
15. General prismoidal correction
16. Homothetic irregular polygonal frustum / correction

Formula reference

This section gives only the general formulae used by the program. No numerical worked examples are included.

Cube

For cube edge length `a`:

$$V = a^3$$

Right rectangular prism / cuboid

For base dimensions `a` and `b`, and perpendicular height `h`:

$$B = ab$$

$$V = Bh = abh$$

Oblique prism / parallelepiped

For base area `B` and perpendicular distance `h` between the parallel bases:

$$V = Bh$$

When the base is rectangular:

$$B = ab$$

$$V = abh$$

The height must be the perpendicular height, not the sloping edge length.

Triangular prism

For triangular base width `b`, triangular height `h`, and prism length `L`:

$$B = \frac{1}{2}bh$$

$$V = BL = \frac{1}{2}bhL$$

Regular hexagonal prism

For regular hexagon side length `s` and prism length `L`:

$$B = \frac{3\sqrt{3}}{2}s^2$$

$$V = BL$$

Cylinder

For radius `r` and height `h`:

$$A = \pi r^2$$

$$V = \pi r^2 h$$

Square pyramid

For square base side `a` and height `h`:

$$B = a^2$$

$$V = \frac{1}{3}Bh = \frac{1}{3}a^2 h$$

As a prismoidal limiting case:

$$A_1 = 0, \quad A_2 = B, \quad A_m = \frac{B}{4}$$

$$V = \frac{h}{6}(A_1 + 4A_m + A_2)$$

Right circular cone

For base radius `r` and height `h`:

$$V = \frac{1}{3}\pi r^2 h$$

As a prismoidal limiting case:

$$A_1 = 0, \quad A_2 = \pi r^2, \quad A_m = \frac{A_2}{4}$$

$$V = \frac{h}{6}(A_1 + 4A_m + A_2)$$

Similar-base pyramid frustum

For lower area `A_1`, upper area `A_2`, and perpendicular height `h`:

$$V = \frac{h}{3}\left(A_1 + \sqrt{A_1 A_2} + A_2\right)$$

The equivalent prismoidal middle area is:

$$A_m = \left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2}\right)^2$$

The same volume may be written as:

$$V = \frac{h}{6}(A_1 + 4A_m + A_2)$$

Conical frustum

For lower radius `R`, upper radius `r`, and height `h`:

$$V = \frac{\pi h}{3}\left(R^2 + Rr + r^2\right)$$

In area form:

$$A_1 = \pi R^2$$

$$A_2 = \pi r^2$$

$$V = \frac{h}{3}\left(A_1 + \sqrt{A_1 A_2} + A_2\right)$$

Triangular wedge

For triangular base width `b`, triangular height `h`, and extrusion length `L`:

$$V = \frac{1}{2}bhL$$

General prismoid / prismatoid

For two parallel end areas `A_1` and `A_2`, middle area `A_m`, and perpendicular spacing `L`:

$$V_P = \frac{L}{6}(A_1 + 4A_m + A_2)$$

This formula requires the actual middle area `A_m`. For arbitrary non-similar irregular end sections, `A_m` cannot be inferred from `A_1` and `A_2` alone.

Sphere

For sphere radius `r`:

$$V = \frac{4}{3}\pi r^3$$

The same result is obtained by Simpson/prismoidal integration across the diameter:

$$A_1 = 0, \quad A_m = \pi r^2, \quad A_2 = 0, \quad L = 2r$$

$$V = \frac{L}{6}(A_1 + 4A_m + A_2)$$

Average end-area method

For two end areas `A_1` and `A_2`, separated by distance `L`:

$$V_{EA} = \frac{L}{2}(A_1 + A_2)$$

This method is exact when the cross-sectional area varies linearly between the two sections. It is generally approximate for prismoidal or frustum geometry unless the geometry specifically satisfies the end-area assumption.

General prismoidal correction

The prismoidal volume is:

$$V_P = \frac{L}{6}(A_1 + 4A_m + A_2)$$

The average end-area volume is:

$$V_{EA} = \frac{L}{2}(A_1 + A_2)$$

With the sign convention used in the script, the correction to subtract from the average end-area volume is:

$$C = V_{EA} - V_P$$

Therefore:

$$C = \frac{L}{3}(A_1 + A_2 - 2A_m)$$

and:

$$V_P = V_{EA} - C$$

Homothetic irregular polygonal frustum / correction

This method applies when:

- the two end polygons have the same plane shape;
- the two polygons lie in parallel planes;
- the two polygons differ only by uniform linear scaling;
- corresponding vertices are joined by straight generatrices;
- `L` is the perpendicular spacing between the planes.

For end areas `A_1` and `A_2`, the middle area is not the arithmetic mean of the end areas. It is the area corresponding to the average linear scale:

$$A_m = \left(\frac{\sqrt{A_1} + \sqrt{A_2}}{2} \right)^2$$

The exact homothetic-frustum volume is:

$$V = \frac{L}{3} \left(A_1 + \sqrt{A_1 A_2} + A_2 \right)$$

The correction to subtract from the average end-area volume is:

$$C = \frac{L}{6} \left(\sqrt{A_2} - \sqrt{A_1} \right)^2$$

Therefore:

$$V = V_{EA} - C$$

This correction is not valid for arbitrary non-similar irregular polygons. If the end polygons have different shapes, different vertex ordering, twist, local distortion, or non-uniform scale change, the real middle area must be constructed, measured, or obtained by a more general geometric or numerical method.

Installation

Requirements

- Python 3.10 or later recommended.
- `numpy`
- `matplotlib`
- Tkinter, normally bundled with standard Python installations on Windows and macOS. On some Linux distributions it must be installed separately through the system package manager.

Windows virtual environment

Open Command Prompt or PowerShell in the project folder:

```
py -m venv .venv
.venv\Scripts\activate
py -m pip install --upgrade pip
py -m pip install matplotlib numpy
```

Linux/macOS virtual environment

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install matplotlib numpy
```

Usage

Place the Python file `script.py` in the project folder.

Run with:

```
python script.py
```

or:

```
python script.py
```

Typical workflow:

1. Select the solid or calculation method from the drop-down list.
2. Enter the required dimensions or areas.
3. Press **Calculate**.
4. Review the calculated volume and correction line.
5. Review the full report in the GUI output box.
6. Use **Save TXT** to write the displayed report to `output.txt`.

The report displayed in the GUI is the same report exported to the text file.

Building a stand-alone executable

Windows PyInstaller one-file build

Inside the activated virtual environment:

```
py -m pip install pyinstaller
pyinstaller --clean --onefile --windowed --name script --collect-all matplotlib
↪  script.py
```

The executable will be created at:

```
dist\script.exe
```

For debugging, replace `--windowed` with `--console` to show console errors. After validation, rebuild with `--windowed`.

Linux/macOS PyInstaller one-file build

```
python -m pip install pyinstaller
pyinstaller --clean --onefile --windowed --name script --collect-all matplotlib
↪  script.py
```

The executable will be created in the `dist` folder.

GUI layout

The interface is intentionally divided into two fixed halves:

- **Left half:** input fields, selected method, calculated volume, correction summary, and full text report.
- **Right half:** large 3D view of the selected solid, with dimension labels and a compact technical plotting style.

The window is non-resizable by design. The dimensions are selected to fit typical laptop and desktop screens while keeping the 3D view readable.

Output files

The script can write the displayed calculation report to:

```
output.txt
```

No database, external data file, or internet access is required.

Notes and limitations

- All length, radius, spacing, and area inputs must be positive.
- The script assumes consistent units. If lengths are in metres and areas are in square metres, volumes are returned in cubic metres.
- For oblique prisms, the height input is the perpendicular distance between bases.
- For prismoids and prismatoids, the middle area `A_m` must be the real middle section area.
- The homothetic irregular-polygon correction is exact only under the stated same-shape, parallel-plane, uniform-scale assumptions.
- The average end-area method is included for comparison and practical use, but it is not generally equivalent to the prismoidal formula.
- The 3D view is a graphical aid for interpretation and does not replace the formula selected for calculation.

Repository contents

Recommended minimal repository structure:

```
.
  README.md
```

```
  script.py
  output.txt        # created when the user saves a report
```

Optional executable build output:

```
.
  dist/
      script.exe
```
